01 / PROBLEM AND CONTRIBUTION

Inverter & NAND standard-cell design

EEE 425/591 · Digital System Circuits · Lab 2 | Fall 2025

RESULT AT A GLANCE

Designed INVx03 and NAND2x04 cells, measured directional propagation delays, and documented clean DRC and passing LVS results for both layouts.

Engineering question

How do transistor sizing and layout verification turn logic gates into checked custom cells?

Situation

A standard cell needs correct logic behavior, characterized transitions, and a layout that matches its schematic. This lab follows two gates through schematic testing and physical verification.

Task

Size and implement INVx03 and NAND2x04 cells, measure their 50% propagation delays, and check their layouts with DRC and LVS.

My contribution

Prepared cell sizing, schematics, symbols, timing plots, layouts, and physical-verification evidence in the submitted report.

Tools and concepts

CMOS standard cells / Transistor sizing / IC layout / DRC/LVS / Timing characterization

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation

Sized the INVx03 with 900 nm NMOS and 1370 nm PMOS widths, and the NAND2x04 with reported widths of 2863 nm and 2260 nm.

02 / Implementation

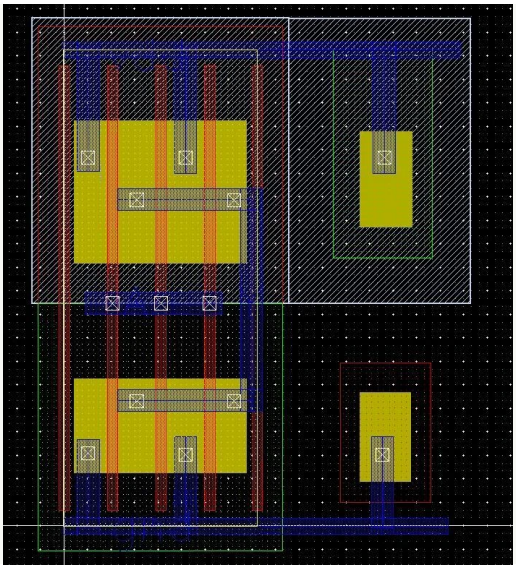
Measured high-to-low and low-to-high propagation delays separately. The inverter values are 11.9 ps and 14.1 ps; the NAND values are 9.31 ps and 16.7 ps.

03 / Implementation

Built both cell layouts and documented the DRC-clean and LVS-pass outputs alongside each implementation.

Timing in two directions

A single averaged delay would hide the asymmetry between pull-up and pull-down paths. Reporting both transitions makes the sizing outcome easier to assess.



INVx03 layout

Source report p. 4: inverter cell layout.

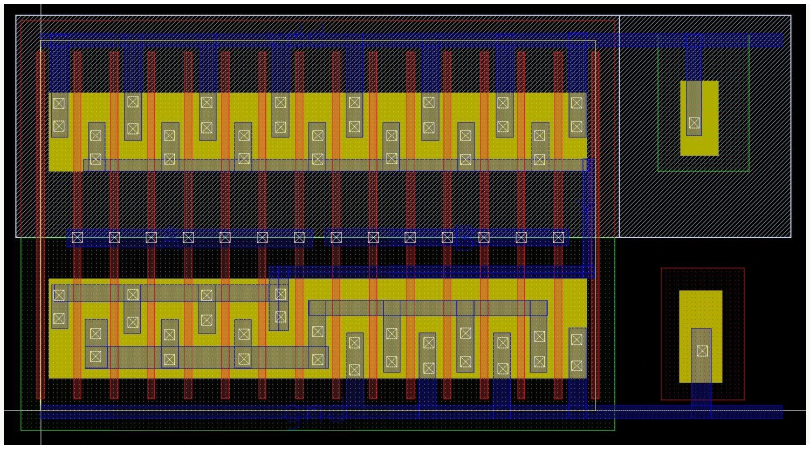
03 / RESULTS AND PERSPECTIVE

Evidence and engineering perspective

INV high-to-low / low-to-high	11.9 / 14.1 ps
NAND high-to-low / low-to-high	9.31 / 16.7 ps
Layout checks	DRC clean · LVS pass

Reported result

Both cell layouts have reported clean DRC and passing LVS results. The directional timing values show unequal rise/fall-path delays, especially for the NAND cell; the case study retains these differences.



NAND2x04 layout

Source report p. 4: two-input NAND cell layout.

Scope and limitations

The listed delays come from the schematic-test section of the report and are not labelled post-layout timing. A complete library characterization across input slew, output load, and process corners is not demonstrated, and ±5% delay symmetry is not claimed.

Possible next steps - proposed work

• Add extracted timing under the same stimulus and output load.

• Characterize slew/load grids and corners before packaging library timing models.

Source material: EEE 425/591 Digital System Circuits Lab 2 submitted report, physical PDF pp. 1-6; delay table p. 3 and verification evidence pp. 5-6.